

===== Smart Cultural Storyteller =====

Objectives:

Most traditional and cultural stories are still shared in text or oral formats, which often feel outdated and less engaging for younger audiences. Creating digital storytelling content usually requires writing scripts, recording narration, designing visuals, and editing videos, which demands both time and technical skills.

The aim of this project is to build an AI-powered system capable of automatically generating complete story videos with audio narration, captions, and background music while keeping authentic, making cultural storytelling more modern, engaging, and accessible.

Methodology:

The system was developed as a complete pipeline using open-source models and frameworks, executed in Google Colab :

1. **Input :** User can either type a prompt or record their voice using the audio option.
 2. **RAG system :** A FAISS-based retrieval system with a corpus of stories ensures cultural grounding and reduces hallucinations and Wikipedia provides fallback for context.
 3. **Story Generation :** Open-source language models generate structured stories. A translation layer ensures models process English internally while users interact in their chosen language.
 4. **Image Generation :** Stable Diffusion models generate styled scene illustrations. Google Colab GPU is used to handle heavy computation.
 5. **Narration & Audio :** Edge TTS provides natural multilingual voices, with gTTS as fallback. Background music is selected using semantic matching of scene descriptions.
 6. **Video Assembly:** Combines scene images, audio narration, background music, and captions into a complete video.
 7. **Gradio UI :** A simple interface allows users to either generate a story from a prompt or paste an existing story to directly generate a video.
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Tools & Technologies Used:

- **Frameworks & Libraries:** Transformers, Sentence-Transformers, FAISS, Deep-Translator, langdetect, MoviePy, Pydub, ffmpeg, Gradio
- **APIs & Services:** Wikipedia API, gTTS
- **Models:** Falcon-H1-0.5B-Instruct, Stable Diffusion XL Base 1.0, Whisper, Edge TTS
- **Environment:** Google Colab

Results & Observations

The system successfully generates multilingual cultural stories and renders them as narrated videos with captions and background music. Two usage modes provide flexibility: story creation from an idea, or video generation from a given story.

- RAG improves coherence, though limitations remain when handling complex cultural contexts.
- Multilingual story generation works, but models still rely heavily on English internally.
- The translation layer bridges this gap, though the translations are not always fully accurate.
- Video generation requires GPUs.

Challenges & Solutions

- **Model Hallucination:** Addressed by integrating a FAISS-based RAG system for factual grounding.
- **Heavy Diffusion Models:** Managed using Google Colab's free GPU resources.
- **English-Centric Models:** Solved with a translation layer for multilingual support.
- **Engagement:** Enhanced through narration, captions, and background music integration.

Conclusion & Future Work

The Smart Cultural Storyteller shows how AI can preserve and present traditional and cultural stories in engaging, multimedia formats. It reduces the complexity of video creation and makes storytelling accessible to anyone. Future directions include fine-tuning models on folk datasets, building a more robust RAG pipeline, and adapting background music to scene moods. Expanding language support, improving rendering speed, and enabling interactive features such as branching storylines could further enhance accessibility and audience engagement.